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Original research article

Proposing a value-sharing mechanism for wind power: Impacts of benefit-sharing and community engagement on acceptance and environmental awareness

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ABSTRACT

As governments increasingly promote installing wind power plants, conflicts with residents have emerged. Some wind farm villages implement financial benefit-sharing schemes and organize public hearings, yet empirical evidence on the effectiveness of such mechanisms remains limited. This study examines whether financial benefit-sharing and community participation are associated with residents' acceptance of wind turbines and with environmental awareness. We propose a Value-sharing Mechanism (VSM) framework by comparing villages with different benefit-sharing types and participation levels. A survey of 95 residents from three wind farm villages in Jeju Island, South Korea, classified as utility-focused, benefit-focused, and participation-focused models, was conducted. The Kruskal-Wallis test, correlation analysis, and linear regression were employed to assess the relationships among perception of benefit-sharing, community engagement, local acceptance, and environmental awareness. Results indicate that both benefit-sharing and community engagement are positively associated with local acceptance, but not with environmental awareness. These findings inform theoretical insights through the lens of commons theory and may inform practice for designing self-governing sustainable renewable energy projects under the VSM framework.

1. Introduction

The South Korean government committed to achieving 30–35% renewable electricity by 2030 and 30–35% by 2040 through the Korea Renewable Energy 3020 Plan [1]. The government has established the Carbon Neutrality Act in 2021 and declared the 2050 Carbon Neutral Strategy to achieve net-zero emissions by 2050 [2]. The government has installed power generation capacity totaling 291,531 GWh, distributed among gas (30.7%), coal (27.6%), renewables (19.8%), and nuclear (17.3%) [3]. Under the 1st National Basic Plan for Carbon Neutrality and Green Growth and the 10th Basic Plan for Electricity Supply and Demand in 2023, the government has set a target to install 19.3 GW of wind power facilities by 2030 and 34.1 GW by 2036 [4].

The Jeju Island's association with wind power development dates back to the mid-1970s, when the first experimental wind turbine was installed. In the 1990s, the government installed commercial-scale wind turbines in Haengwon-ri, marking the first significant deployment of wind energy for electricity generation on the island. Since then, large-scale onshore and offshore wind farm projects have been installed in line with the Carbon Free Island Jeju 2030 initiative [5]. Under the Wind Power Generation Permission Special Law, a public-led investment promotion plan for wind power development was unveiled in September 2015. Subsequently, the Renewable Energy Specialized Village System promoted small-scale (3 MW) wind power projects led by residents.

However, concerns raised by residents because of noise and landscape alterations [6]. The community engagement in wind farms exist in multiple local contexts in South Korea [7–9]. In general, conflict mechanisms can exacerbate social opposition and perceptions of renewable energy projects, which are often driven by external governmental financial aid [10], or the distribution of financial benefits [11]. This highlights the need for mechanisms of technological innovation and social consensus [12].

The benefit-sharing mechanism (BSM) has been studied as a conceptual framework to understand systemic interactions between power

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plant operations and residents based on a community development approach. A systematic review of social opposition and renewable energy stakeholders revealed that the lack of community engagement during the decision-making process is a primary reason for opposition [13]. Previous studies have suggested frameworks for the BSM based on financial benefits and community engagement and a resident-led power plant model has been proposed [14–16]. In South Korea, a resident-led wind farm model was first introduced on Jeju Island in 2016 [11].

However, studies on perceptions of community-based renewable energy projects are scarce [17]. Existing studies on wind turbines primarily focus on public opinion regarding their installation [8,9]. However, there remains a significant gap in research on benefit-sharing mechanisms. Only a few empirical studies have examined benefit-sharing in the context of onshore wind farms in South Korea, and most of these are based on interviews. Moreover, external factors such as environmental awareness have not been sufficiently explored in relation to the influence of benefit-sharing mechanisms.

This study seeks to answer the following research questions through empirical analysis, focusing on local acceptance and environmental awareness. Specifically, this study focuses on following questions: 1) Are there differences among the utility-focused, benefit-focused, and participation-focused village models in their local acceptance and environmental awareness? 2) Is the perception of benefit-sharing positively associated with local acceptance and environmental awareness? 3) Is community engagement positively associated with local acceptance and environmental awareness?

With the lens of commons theory, this study investigated three wind farms where installed wind turbines on the shared pastureland and received funding from wind power generation companies, which are utility-focused (Samdal-ri), benefit-focused (Gasi-ri), and participation-focused (Haengwon-ri) village models in Jeju Island. The sites were selected because of their similarity in size of wind turbine installation and disparity in benefit-sharing framework presented in Musall and Kuik [15]. Based on the results, we propose a novel approach to the framework of a value-sharing mechanism (VSM) instead of the traditional benefit-sharing mechanism (BSM). The VSM encompasses a broad range of value-sharing aspects, including financial benefits, community engagement, and environmental awareness. These findings offer theoretical insights through the lens of commons theory and practical implications for designing self-governing, sustainable renewable energy projects under the VSM framework.

Section 2 discusses the theoretical and conceptual framework, focusing on commons theory and the value-sharing mechanism. Section 3 outlines the research design, including the research questions, case study description, data collection methods, and analytical approach. Section 4 presents the results, while Section 5 offers a discussion of the findings. Finally, Section 6 concludes with recommendations for future studies.

2. Theoretical and conceptual framework

2.1. Commons theory

Since community renewables involve issues related to shared land property and natural resources, this study is framed within the context of the commons theory. This theory began with Hardin's "The Tragedy of the Commons," which describes the problems of overuse and degradation of natural resources [18]. Elinor Ostrom's book Governing the Commons [19] and her article [20] proposed eight design principles for the governance of common-pool resources: (1) clearly defined boundaries; (2) rules tailored to local circumstances; (3) participatory decision-making; (4) monitoring; (5) graduated sanctions for those who abuse the commons; (6) accessible conflict resolution; (7) the community's right to self-organize; and (8) nested, larger networks [19,21]. Since then, Cox et al. [22] has applied the commons theory to managing natural common-pool resources, such as fisheries and forests. Furthermore, Bauwens et al. [23] applied the commons framework to renewable energy and examined the factors that foster community engagement in wind power cooperatives in Denmark, Germany, Belgium, and the United Kingdom, while Wade and Ellis [24] used wind rent in conjunction with Ostrom's institutional analysis. Andersson [25] employed the Institutional Analysis and Development (IAD) framework to identify which institutional factors influence the likelihood of successful governance.

This research aligns with Bauwens et al. [23] and Wade and Ellis [24], which view wind as a common-pool resource. We analyze three cases of Jeju Island in South Korea, where commercial wind turbines are installed on shared pastureland managed by village committees. Led by the government or commercial wind power companies, compensation is provided to the village committees, which then manage the community funds. These case studies are consistent with the commons theory described in Ostrom [21], as they involve clearly defined village boundaries, community-specific rules, local decision-making, and the right to self-organize. This research explores the institutional and socioeconomic factors that influence the likelihood of managing wind farms through the lens of commons theory.

2.2. Conceptual theory of benefit sharing mechanism

The benefit sharing mechanism (BSM) is conceptualized as a system that distributes direct and indirect benefits derived from power generation projects among stakeholders or local communities, given diverse terms such as "benefit payment" or "compensation to the community". Comparative analyses have further explored the distinctions between corporate-led and resident-driven power generation models, contributing to a nuanced understanding of BSM dynamics [15]. Studies have explored diverse aspects, including resident attitudes toward wind power projects, barriers to benefit sharing, and the role of BSM in enhancing local acceptability [16].

Since community-engaged renewable energy became more prevalent in the early 2000s, the BSMs began to be studied in the United Kingdom, especially in Scotland [16,26–29]. Following these studies, more countries have adopted community-based renewable energy generation, and research has been conducted primarily in European countries such as Sweden [30], France [31], Denmark [32,33], Germany [34], and the Netherlands [35], while some studies have also examined community-based approaches to renewable energy projects globally, such as in Tunisia [37] and Australia [36]. South Korea has also been exploring the benefit-sharing approach since the late 2010s [11,38].

The BSM has been examined in relation to local acceptance of renewable energy projects [37]. Prior work spans specific technologies on wind power [28,39], solar power [40,41], and hydropower [42], as well as mixed wind–solar contexts where community engagement is salient [43–45]. Research has also examined differences in local acceptance of offshore wind and onshore wind [46,47]. In the literature, community engagement is sometimes used broadly to include both participation in decision-making and, in some contexts, financial participation through community ownership. In this paper, we retain this broad conceptual usage in the review. For the empirical analysis, however, we interpret community engagement in the decision-making sense (e.g., public hearings, consultations, committee processes); details of the operational measure are provided in Section 2.2.2.

The BSM has mainly been studied through two approaches: financial benefits and residential ownership schemes [49], which suggest community benefit-sharing and renewable energy projects involving granting access to electricity, community benefit funds, and community ownership.

First, regarding financial benefits, a community-based approach describes how residents receive and distribute financial benefits from renewable energy projects [14,30,47,48]. Usually, the benefit-sharing in renewable energy is directed to community funds. For example, in the United States, benefit-sharing has been distributed through community

fund-based systems [35]. The concept of funding has also been referred to as community funds in Sweden [30], and fund allocation schemes in Scotland [28]. In Scotland, funding allocation encompasses not only cash flow but also community benefit agreements [29].

Second, with respect to the ownership schemes in renewable energy, studies on residents' community ownership (including co-ownership) examine developer–community dynamics [14] and how community benefits are apportioned and delivered [16]. Empirical work reports that community ownership is associated with higher local acceptance in Germany [15] and Scotland [44], and qualitative interviews in South Korea suggest similar patterns [11,38]. Conceptually, community ownership is distinct from BSM: ownership embeds residents in the project's governance and risk–return structure (e.g., equity and voting rights), whereas BSM primarily concerns the distribution of project benefits (e.g., community funds or agreements) without implying ownership.

This study examines the self-governance of village committees that manage wind turbines installed on shared pasture. We focus on the associations among benefit-sharing, community engagement, local residents' acceptance of wind turbines, and environmental awareness, as detailed in the following sections.

2.2.1. Benefit-sharing

Benefit-sharing can take various forms, including direct or indirect investment, project financing, scholarship funding, educational programs, energy price reductions, village fund formation, local employment, benefit distribution, development fund establishment, shared ownership, community tax refunds, and electricity tariff reductions. In this study, benefit-sharing is categorized into the following types: (1) village funds provided by developers to residents, often managed by village institutions, local governments, or community organizations, and used for infrastructure or vulnerable groups [51,52]; (2) compensation for potential or realized damages [53]; (3) benefits-in-kind, such as non-financial support including food or educational programs [54]; (4) local employment during the development, construction, and operation [41]; (5) energy price reductions, offering discounted electricity rates to residents [55,56]; and (6) indirect social benefits, such as tourism reputation effects [16,54]. Based on prior work, village funds, compensation, local employment, and energy price reductions are among the most frequently reported types.

In this paper, the term compensation is used in a broad sense to encompass direct and indirect benefits provided to community members. When used in a narrow sense, it is limited to cash payments from companies to residents as direct compensation for damages. Where we refer specifically to this narrow usage, we explicitly indicate "direct cash compensation."

2.2.2. Community engagement

Community engagement in wind projects refers to how project developers or public entities involve community members in decision-making processes (e.g., information, consultation, committee-based deliberation) [56,60]. Related arrangements, such as benefit-sharing (e.g., community funds or agreements) and community ownership (equity stakes), may co-occur but are treated as distinct concepts in this paper. At the global level, many wind and other renewable energy projects are developed and owned by companies; community engagement in such projects concerns how local residents are involved in governance and how benefits are distributed, rather than implying resident-only development [57].

The degree and form of engagement vary across mechanisms (e.g., information disclosure and consultation, community consultative committees, and community benefit agreements or funds) and across project phases (planning, development, operation) [57,60]. In Australia, Howard [57] proposed a framework highlighting consultative committees and contributions to local infrastructure. Similarly, Action Renewables [49] and Lee and Yun [60] discuss graduated stages of community participation in governance and, separately, opportunities for community ownership; ownership opportunities have been linked to higher local acceptance.

Community engagement in wind projects refers to arrangements that embed local participation in decision-making and benefit-sharing and, in some contexts, co-ownership or equity, regardless of whether the primary developer is a company, a public entity, or a resident-led organization [57,60]. At the global level, many wind and other renewable energy projects are developed and/or owned by companies; community engagement in such projects concerns how local residents are involved in governance and how benefits are distributed, rather than implying resident-only development [57].

What is often termed "community wind" may involve various degrees of resident ownership and operation (including co-operative or shared-ownership models), sometimes referred to as community power [58]. For example, Denmark mandates that up to 20% benefit-sharing by offering 20% of project equity be offered to nearby residents living within 4.5 km of a wind farm; this ownership/benefit-sharing mechanism has been reported to be associated with higher public acceptance and is frequently cited as supporting the expansion of wind energy in the country [33,59]. In South Korea, public dialogue has been highlighted as a critical transition strategy to mitigate community opposition to wind development [5].

In line with Lee and Yun [60], this study adopts a five-stage classification of community engagement that reflects increasing resident participation in decision-making and opportunities for community equity or private investment. In the Korean context, typical forms of engagement include public hearings or consultations convened by village committees, wind power companies, or local governments during planning and approval processes, as well as shared-ownership arrangements in which a village committee may hold an equity stake in a wind power plant [60]. In the empirical analysis that follows, we interpret community engagement in the decision-making sense only; ownership or equity is treated as a related but distinct construct and is not included in the operational measure.

2.2.3. Local acceptance

In this study, local acceptance refers to the degree of residents' perceptions and their individual willingness to accept wind power generation in their village. Low local acceptance often stems from concerns about limited job opportunities, negative impacts on the landscape, health risks, delayed decision-making processes, and environmental concerns [61]. Low local acceptance of onshore wind projects can increase the risk of project failure, lead to higher costs, and delay implementation, underscoring the importance of enhancing local acceptance [31].

In prior studies, local acceptance has been examined in relation to the Not-In-My-Back-Yard (NIMBY) phenomenon [62–64]. In South Korea, exploratory work have examined social conflict and community engagement around wind power [65] and solar energy projects [45]. However, survey-based evidence on how financial benefit-sharing is associated with local acceptance of wind turbines remains limited.

2.2.4. Environmental awareness

Environmental awareness is closely linked to local acceptance in renewable energy projects, yet it has been less examined in the context of benefit-sharing models. In practice, wind projects may generate perceived local environmental externalities, such as landscape change, noise, and risks to birds and wildlife, despite their broader contributions to climate and air-quality goals [39]. Qualitative research studies on environmental awareness typically analyze respondents' demographic characteristics and attitudinal patterns, showing mixed associations [65]. By contrast, few survey-based empirical studies have investigated whether community engagement or benefit-sharing is associated with residents' environmental awareness. This study addresses that gap by examining perceived benefit-sharing and perceived community

engagement alongside local acceptance and environmental awareness.

3. Research design

3.1. Research questions/hypotheses

Guided by the research framework, this study formulated research questions and designed a survey to examine whether benefit-sharing and residents' participation are associated with local acceptance of wind turbine installation and environmental awareness. As shown in Fig. 1, we define the variables as follows. Benefit-sharing (X_1) is measured by respondents' perceptions of benefit-sharing in their village (survey item A3; 5-point Likert, higher values indicate stronger perceived benefit-sharing). Community engagement (X_2) is measured by respondents' perceptions of decision-making participation at the village level (survey item A5; 5-point Likert, higher values indicate greater perceived engagement). Ownership or equity is not included in this measure. Local acceptance (Y_1) reflects local residents' favorable attitudes toward wind turbine installation, and environmental awareness (Y_2) reflects residents' awareness regarding environmental issues. The regression models are used to examine the strength and sign of associations rather than to identify causal effects.

With this conceptualization, the research questions were structured as follows (See Fig. 1).

Building on this framework, we pose the following research questions and hypotheses:

Research Question 1: Do villages that have adopted different participation/benefit-sharing arrangements—Samdal-ri (utility-focused), Gasi-ri (benefit-focused), and Haengwon-ri (participation-focused)—differ in local acceptance and environmental awareness? Based on the three village–arrangement groups defined in RQ1, we test the following.

Hypothesis 1-1. Local acceptance differs across villages.

Hypothesis 1-2. Environmental awareness differs across villages.

Research Question 2: Are residents' perceptions of benefit-sharing positively associated with local acceptance and environmental awareness?

Hypothesis 2-1. Perceived benefit-sharing (X_1) is positively associated with local acceptance (Y_1).

Hypothesis 2-2. Perceived benefit-sharing (X_1) is positively associated with environmental awareness (Y_2).

Research Question 3: Are residents' perceived community engagement positively associated with local acceptance and environmental awareness?

Hypothesis 3–1. Perceived community engagement (X_2) is positively associated with local acceptance (Y_1).

Hypothesis 3–2. Perceived community engagement (X_2) is positively associated with environmental awareness (Y_2).

3.2. Case study description

Drawing on field investigations and in-depth interviews, the case studies show that wind turbines were installed on shared pasture. Particularly, the wind power sites in Haengwon-ri, Samdal-ri, and Gasi-ri were originally village communal pastures, which are regarded as common-pool resources collectively owned and managed by residents. Thus, decisions to install wind power projects on the shared pasture required local resident's agreement and their self-governance based on funds provided by the wind power companies.

The village committee is composed of residents recognized as full members of the community, a status that in Jeju Island's long-established villages is generally reserved for long-term inhabitants. Newly relocated individuals, even if officially residing within the village's administrative boundaries, often require a significant period before being regarded as full members eligible to serve on the committee. As a result, committee members are often long-time residents who grew up in the village, most of whom are farmers over the age of 60. The village committee determines the types of benefit-sharing from wind power companies and how these benefits are allocated to members. This institutional context aligns with this study's focus on village-level management of benefit sharing and community engagement.

Therefore, this study identified Haengwon-ri, Samdal-ri, and Gasi-ri as the research sites (Fig. 2). Each village hosts at least ten onshore wind turbines and operates independently within its local context, offering a suitable basis for comparative analysis of their respective benefit-sharing mechanisms.

3.2.1. Samdal-ri

Geographically unified as Samdal-ri, administratively divided into Samdal1-ri and Samdal2-ri, each operates independently since 2019. The Samdal-ri onshore wind power generation site has been led by a private wind company. Out of the 11 installed wind turbines, 6 were erected on privately owned land acquired by the companies and 5 were installed on the shared pasture of Samdal-ri since 2009. The private wind company provided funding to the Samdal-ri village committee as compensation for land use, and the village committee received it as a village committee fund.

Regarding the community engagement scheme, the role of village committee is mostly focused on the allocation of funding. Through the decision-making process of the village committee, the fund was allocated between Samdal1-ri and Samdal2-ri in a 7:3 ratio, based on the

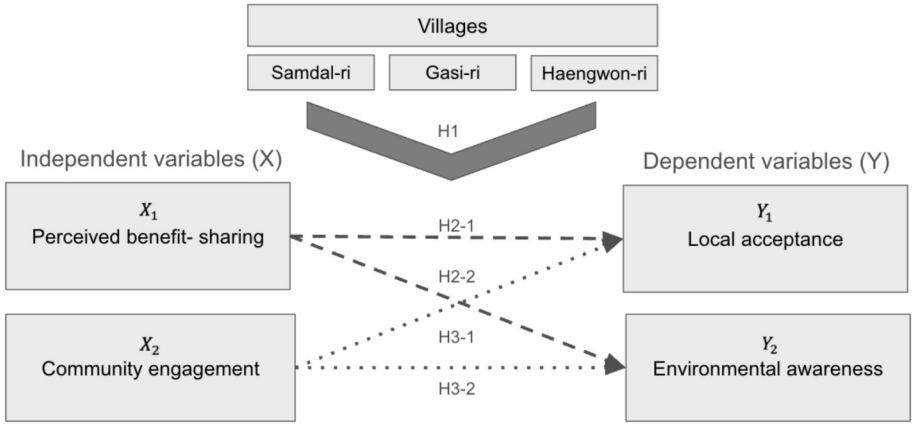

Fig. 1. Research design.

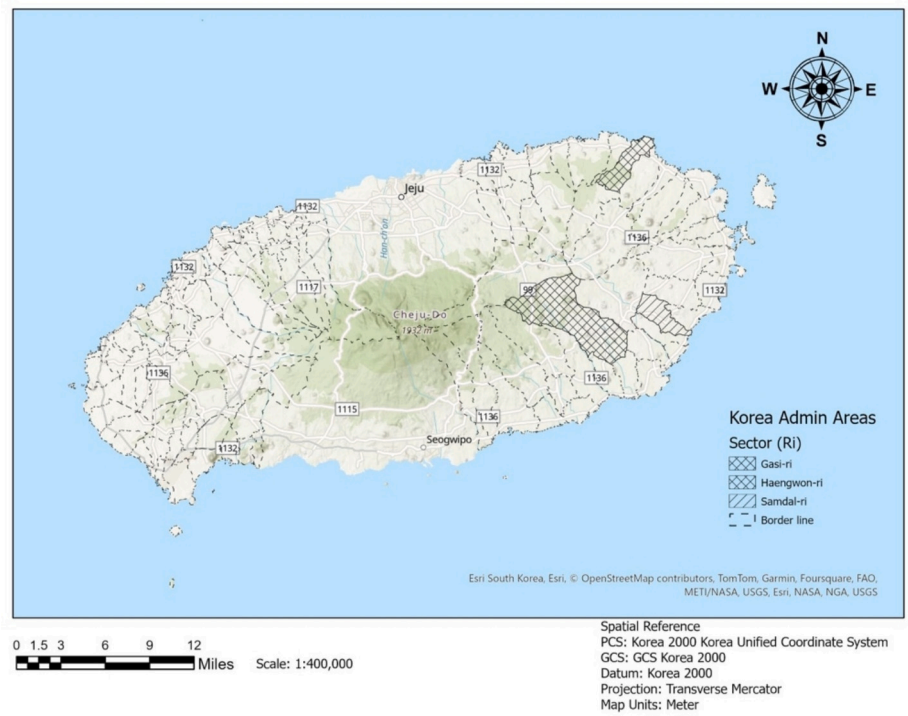

Fig. 2. Study site.

number of village committee members in each area.

For the benefit-sharing scheme, each village committee used most of the funds for village operation, such as rebuilding the village hall, providing scholarships to students and food support during holidays. However, challenges arose due to the administrative division between Samdal1-ri and Samdal2-ri, and wind power company's delayed payment of land rent to village committee, due to tax payment, which makes difficult for the village to secure funds.

3.2.2. Gasi-ri

In 2010, Gasi-ri village initiated a land-based wind power and solar power generation project, receiving funding annually for the power plant companies to use the village committee's joint pastureland, and allocated the funding to a village committee fund. Initially, a contract was established between the village and the Jeju Special Self-Governing Provincial Office (Jeju Provincial Office), but the village later had a direct contract with the Jeju Energy Corporation on behalf of the Jeju Provincial Office. Subsequently, a private wind power company installed ten wind turbines with a total capacity of 15 MW under a 15-year contract.

Under the leadership of the village committee, Gasi-ri has pursued various forms of benefit-sharing with residents, positively influencing perceptions. Funds generated from wind power projects are allocated to village committee councils, which then distribute them to the village operation for communal purposes. While there are no regular meetings to gather residents' opinions, decisions were made by the village committee leader, who later announced how the funding would be allocated to various communal initiatives.

Regarding community participation, the initial wind power generation project introduced by the Jeju Provincial Office was followed by Gasi-ri. However, during the signing of the 15-year contract, the rent was set at a fixed rate. Challenges arose in integrating benefit sharing with fluctuating land prices. As land prices typically rise, necessitating lease fee adjustments, the financial gains from wind power generation remain fixed, resulting in a decline in real benefit value over time. Discontent within the village has emerged due to the lack of reflection on changes occurring during the long-term contract period, prompting plans for intense negotiations upon contract expiration to renegotiate benefits.

3.2.3. Haengwon-ri

Haengwon-ri employs a combination of offshore and onshore wind power generation, comprising 15 wind turbines, one offshore turbine managed by the Jeju Provincial Office, and fourteen turbines operated by the Jeju Energy Corporation. Upon the expiration of the 20-year contract period, 5 turbines were dismantled as of 2024. Additionally, Haengwon-ri boasts one wind turbine jointly owned by village cooperative members and the wind power company, reflecting a 50% corporate and 50% resident partnership, known as a resident-participated wind turbine. Haengwon-ri is a special case that installed the resident-led wind turbine with support of Jeju City government. The Haengwon-ri village committee received funds from the wind power company to cover the installation costs of one wind turbine.

While most of the fund is used for the village operation, there were many stakeholders to increase community engagement during the installation of turbines. The wind power company supports the operation of the village corporation and the management of the wind turbine. Haengwon Wind Power Special Purpose Corporation (SPC) is responsible for settling the revenue from the electricity generated by the wind turbine, paying the contract amount to the wind power company, and providing 8% of the electricity sales revenue to the village association as the village committee fund. The investment cost for the project was initially covered by the wind power company, and the contract between the SPC and the wind power company was established on a deferred payment and installment basis. After all the contract payments are settled, it is anticipated that the wind turbines are expected to generate an income annually, and the Jeju Energy Corporation offers financial support of wind turbine revenue to the village committee.

There were network of stakeholders to work with village committee during the installation and operation of the wind turbines. First, the Jeju Energy Corporation and the wind power company conducted explanatory community meetings, such as public hearings, to ensure the satisfaction of village residents and to exchange opinions during the decision-making process. Additionally, there were educational support

programs for village students through after-school classes, and multiple consultation meetings with the village committee. Second, since the village encounters legal, tax, and advisory support challenges, the Jeju City not only supported wind turbine permits and deliberation for companies but also prioritized community engagement of the SPC shareholder. Jeju City was responsible for wind turbine permits and deliberations. Third, the SPC comprises 5 village representatives including village committee leader and key stakeholders in wind power generation.

As a result, the wind turbine installation process initially faced resident opposition due to top-down government. However, local acceptance increase was observed after increasing community engagement.

3.3. Status of benefit sharing mechanism

3.3.1. Comparison of benefit sharing scheme

This study categorized the existing forms of benefit-sharing into the benefit-sharing types based on the previous studies [49,54]. In this study, compensation refers specifically to cash payments, distinguishing it from benefits-in-kind, which are provided in non-cash forms. For example, scholarships for students, considered a form of cash support, are categorized under compensation, while food (rice) support provided to residents on holidays is categorized under benefits-in-kind. Comparing the three villages, the study acknowledges that each village has a different benefit-sharing structure (See Table 1).

There are two stages of benefit-sharing: (1) the first stage is financial compensation given by the wind power company to the village committee as village community funds, which are the same for Samdal-ri, Gasi-ri, and Haengwon-ri; (2) the second stage is the sharing of financial benefits given by the village committee to residents or members, which are decided by the past and current village committees and depend on village leadership, showing different benefit-sharing schemes.

While the common benefits classified based on previous studies, such as village funds, are the same for all villages, in this study, we decided to disregard the impact of benefit types such as local employment and indirect benefits from increasing tourism, since there were no local residents hired by the wind power companies and it was difficult to measure the increase in tourists due to the wind power projects, which was only identified through residents' interviews. We also disregarded cash support from the village committee to residents, as scholarships were only given to specific students and were not widely perceived by residents, and food support was only provided on holidays in all villages.

However, the distinctive benefit-sharing decided by the village committee and given to residents includes the electricity bill subsidy in Gasi-ri and the shared ownership of a resident-led wind power plant in Haengwon-ri, which make the difference among the villages. An exceptional case in Gasi-ri involved allocating energy cost support, such as electricity bill subsidies, to residents. Since all funds provided by wind power generation companies are pooled with community funds under the village committee's financial management, the cash and food support distributed by the village committee are not directly recognized by residents as benefits from wind turbines. However, residents who received electricity bill subsidies from the village committee acknowledged that these benefits were provided after the installation of wind turbines in the village. In contrast, Haengwon-ri granted residents ownership rights over wind turbines, reflecting a 50% corporate–50% resident partnership. The village committee received not only one-time funding support from wind power generation companies but also annual net income from the resident-led wind power generator. These incomes are included in the community fund and used for communal purposes, such as renovating the village committee building. But there was no direct financial benefit distribution from village committee to residents such as electricity bill subsidy in Gasi-ri.

3.3.2. Comparison of community engagement scheme

Based on information from the three villages, the level of community engagement was found to vary in scale. Figs. 3–5 illustrate the stakeholder networks involved in value-sharing in Samdal-ri, Gasi-ri, and Haengwon-ri. A notable observation is that each village has a different number of stakeholders and varying degrees of network complexity. For instance, Samdal-ri features a simple and direct hierarchy for distributing financial benefits, with the wind power company providing funds to the village committee, which then decides how the funds are used. In Gasi-ri, both a wind farm committee and a village committee are involved in distributing funds to residents and wind farm members. Haengwon-ri has the most complex structure, with a central decision-making body composed of the village committee and a SPC, which manages funds from the wind power company. Haengwon-ri also held numerous public hearings during the initial stages of wind turbines installation.

Based on the framework of Lee and Yun [60], which illustrates the five stages of private investment and community engagement percentages classified by ownership types and descriptions, this study classified Samdal-ri, Gasi-ri, and Haengwon-ri into stages of community

Table 1
Status of benefit-sharing in villages.

Benefit-sharing				Villages		
Stage	Stakeholder	Types		Samdal-ri	Gasi-ri	Haengwon-ri
1	Benefits given by the wind power company to the village committee	General/ Shared benefits	Village fund	Community funds for the village	Community funds for the village (Community ranch maintenance)	Community funds for the village
			Local employment	None	None	None
			Indirect benefit	None	Increased tourism driven by the promotion of eco-friendly village image	Increased tourism driven by the promotion of eco-friendly village image
2	Benefits given by the village committee to residents/ members	General/ Shared benefits	Compensation	Cash support (Scholarships for specific students)	Cash support (Scholarships for specific students)	None
			Benefits-in kind	Food support on holidays	Food support on holidays	Food support on holidays
		Village-specific benefits	Energy price discount	None	Energy cost support (electricity bill subsidy, and cable broadcasting fee support)	None
			Local ownership	None	None	Owns one resident-led wind power generator

Note: Classification of benefit-sharing types is based on Action Renewables et al. (2015) [49], and Allan et al. (2011) [54]. Identification of benefit-sharing is based on interviews and field studies in each village.

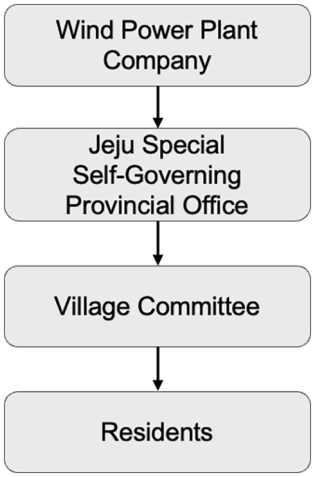

Fig. 3. Samdal-ri Benefit-sharing Mechanism Stakeholder
Note: Visualized based on interviews.

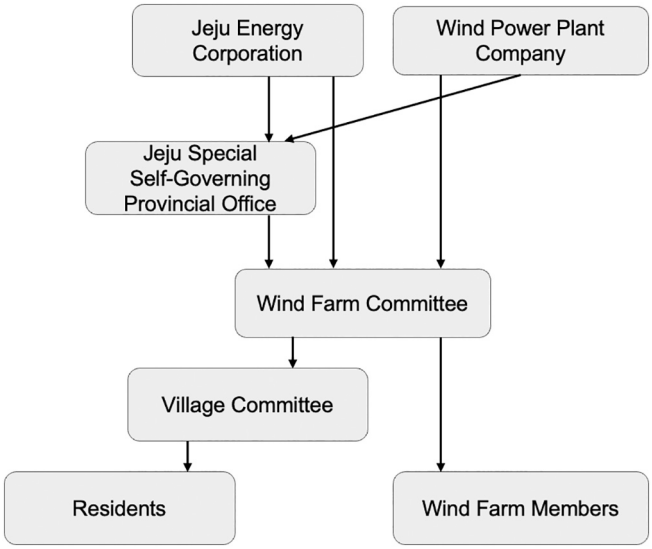

Fig. 4. Gasi-ri Benefit-sharing Mechanism Stakeholder
Note: Visualized based on interviews.

engagement based on interviews and field studies from a previous study [11], as shown in Table 2. Samdal-ri is classified as Stage 1, since the village committee participated by signing the consent form as a land lessor but did not hold any public hearings. Gasi-ri is classified as Stage 2, as the village committee voluntarily shared opinions with the wind farm committee without holding official public hearings, but received benefits from local governments and the company that go beyond residents' consent. Haengwon-ri is classified as Stage 3, since the village committee holds a 50% share of ownership in one wind power project as part of the Haengwon Wind Power SPC, holding multiple public hearings with the Jeju Energy Corporation for residents and corporation committee members to promote residents' understanding of the purpose of building a resident-led wind power plant for the purpose of operating a renewable energy project aligned with Jeju City's initiative, and how financial benefits will be received and distributed to residents.

3.3.3. *Classification of benefit sharing mechanisms*

Walker and Devine-Wright [66] defined the communities of localities with geographical boundary of ownership. Berka and Creamer [67] reviewed community-owned renewable energy projects and found that while empowerment and access to affordable energy remain underexplored, local impacts are more likely realized through indirect, long-term outcomes such as the investment of project revenues within the community. In this study, we used this framework based on benefit sharing and community ownership schemes.

Fig. 6 shows the modified framework based on Musall and Kuik [15] and Walker and Devine-Wright [66]. The X-axis delineates the scope of benefit sharing, while the Y-axis denotes resident participation during wind turbine installation in the village. The X-axis outcome represents financial benefit-sharing centers. When it is more distant and private, the position is regarded as industry-centered benefit sharing. Conversely, when it is more local and collective, the value-sharing status is considered resident-centered benefit sharing. The Y-axis represents the process through the number of meetings between the village committee, residents, and the wind power company, also known as public hearings.

Regarding financial benefit-sharing scheme, Samdal-ri initially received minimal financial benefits during wind turbine installation but needed distinct benefit-sharing modalities beyond community funds and public hearings, typical of common wind farm areas. Gasi-ri village committee managed its benefit-sharing initiatives, distributing electricity support to residents. Haengwon-ri, on the other hand, featured numerous public hearings, a wind turbine operated by private wind power companies, and a SPC comprising five village committee members. Consequently, Samdal-ri denotes the traditional wind farm model,

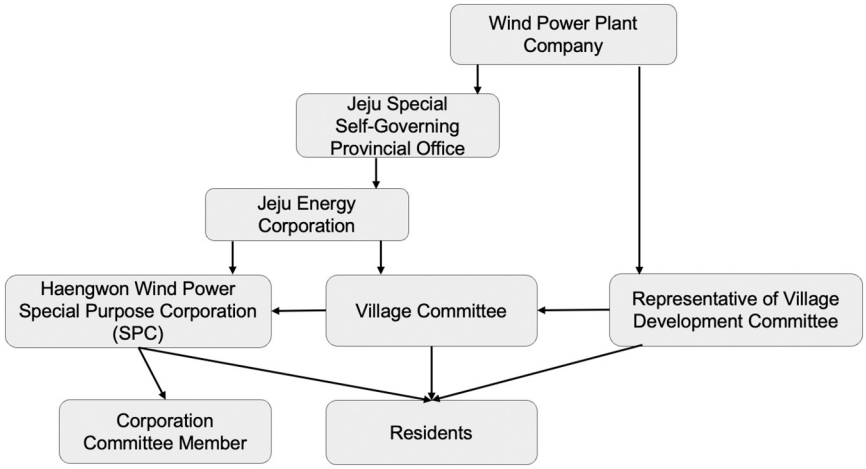

Fig. 5. Haengwon-ri Benefit-sharing Mechanism Stakeholder.
Note: Visualized based on interviews.

Table 2
Stages of community engagement in villages.

Stage	Private investment	Community engagement	Ownership Type	Description	Case
1	100%	0%	Commercial Ownership	Participation in consent or related procedures as required by institutional or administrative regulations; in some cases, a small amount of compensation or support may be provided.	Samdal-ri
2	100%	0%	Commercial Ownership	The village committee, as a land lessor, plays a major role in opinion gathering and public hearings.	Gasi-ri
3	Over 50%	Under 50%	Joint Ownership	The village committee representing residents holds a share of ownership in the wind power project.	Haengwon-ri
4	Under 50%	Over 50%	Co-development or Joint Venture	The village committee played a leading role from the development stage, participating in financing and decision-making processes such as opinion collection.	
5	0%	100%	Community-led Development and Ownership	The village committee independently owns and operates the wind turbines from installation to operation.	

Source: Modified from Lee and Yun [60]. Classification of villages is based on interviews and field studies in each village.

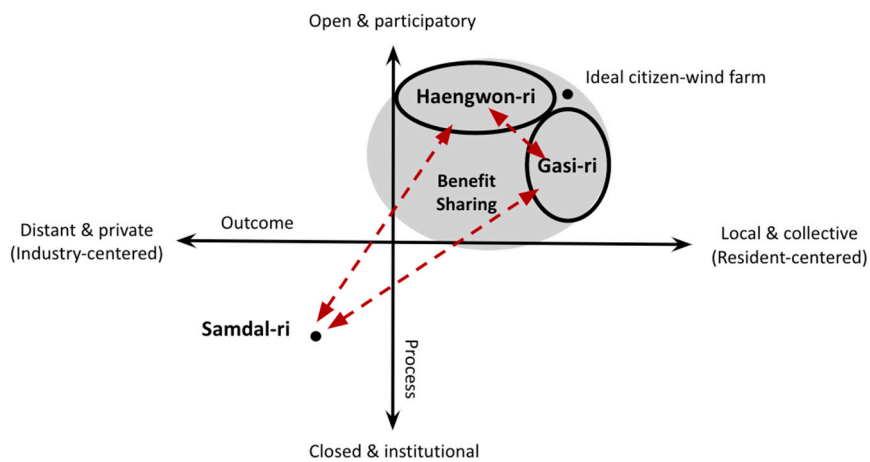

Fig. 6. Modified theoretical framework for the benefit-sharing mechanism.
Source: Modified from Fabian David Musall and Onno Kuik (2011) [15], Gordon Walker and Patrick Devine-Wright (2008) [66].

characterized by a closed and institutional process during wind turbine installation. Gasi-ri depicts the benefit-focused model, emphasizing resident-centered benefit sharing. Haengwon-ri classifies as participation-focused sharing model due to numerous village meetings including public hearings.

In Hypotheses 1–1 and 1–2, the term *villages* refers to (1) voluntary electricity price support within the community organized by the village committee (Gasi-ri), (2) ownership of resident-led wind power projects and public hearings facilitated by the village committee (Haengwon-ri), and (3) conventional company-led wind farms without either benefit-sharing or community ownership (Samdal-ri).

3.4. Data collection methods

This study employed a mixed-methods research approach, starting with an exploratory qualitative phase to identify appropriate case study

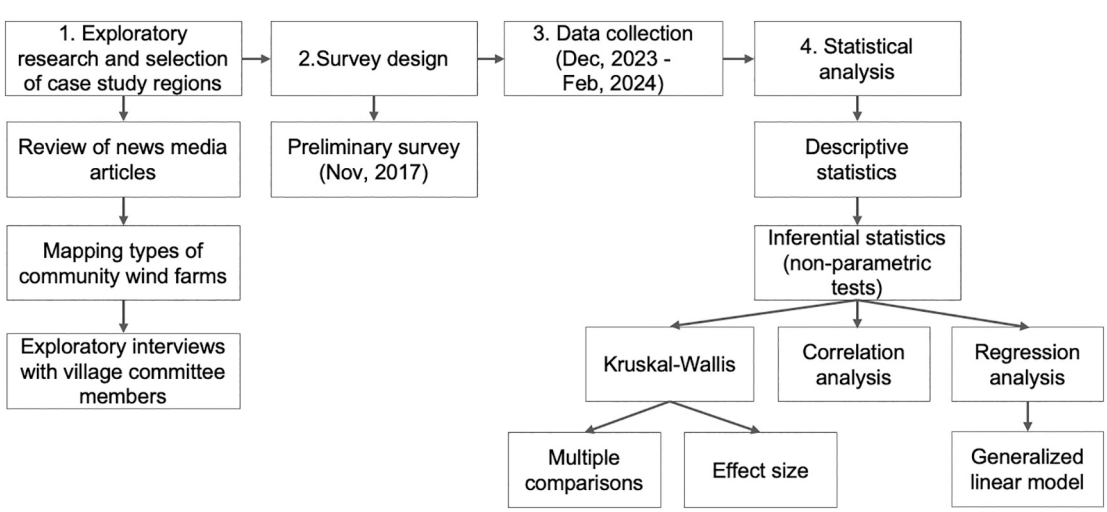

Fig. 7. Flow chart of the methodology.

regions, followed using surveys. As shown in Fig. 7, we conducted exploratory research and selected case study regions by reviewing news media articles and mapping the types of community wind farms. We also conducted exploratory interviews with village committee leaders when explanation of situations is needed.

The first preliminary survey was conducted in November 1–21, 2017, and the second survey was conducted from December 8 to February 23, 2024. Before conducting the second survey, Institutional Review Board (IRB) approval was obtained (IRB number: P01–202311–01-003). In the first survey, 90 questionnaires were gathered. In the second survey, 100 questionnaires were collected, with a minimum of 30 respondents from each village. Excluding 5 respondents who did not meet the survey criteria, 95 questionnaires were used for data analysis.

The survey results were analyzed using Python, providing descriptive and inferential statistics through Kruskal-Wallis tests, correlation,

Table 3
Survey questions and scales.

Classifications	Measurement	Codes	Questions	Source	Scales
Level of resident's perception about benefit-sharing and community engagement	Perception of benefit-sharing status	A1	Did you know about the current benefit-sharing status for wind turbines?	Lee and Yun (2018) [11]	Nominal
		A2	What kinds of benefits did you receive after the wind turbine installation?		Nominal
	Satisfaction of benefit-sharing status	A3	How satisfied are you with the current benefit-sharing from the wind turbine?		Likert 5 scale
		A4	What is your preferred benefit sharing type for future wind turbine installations?		Nominal
	Satisfaction of community engagement	A5	How satisfied are you with the community engagement from the wind turbine? (If you don't experience community engagement during decision-making process in your village, what is your preferred level of participation in future wind turbine installations?)		Likert 5 scale
		A6	What are your preferred types of participation in future wind turbine installations?		Nominal
Level of resident's acceptance about wind turbines	Attitudes toward wind turbines	B7	Did you agree with the initial/first wind turbine installation?	Park and Hwang (2016) [68]	Nominal
		B8	Do you currently perceive the existing wind turbine positively?		Nominal
	Wind turbine advantage	B9–1	Do you think the wind turbine improves air quality?	Musall and Kuik (2011) [15]	Likert 5 scale
		B9–2	Do you think wind energy is an infinite resource?		Likert 5 scale
		B9–3	Do you think the wind turbine enables this island to generate and use energy independently?		Likert 5 scale
		B9–4	Do you think the wind turbine brings economic benefits to the village?		Likert 5 scale
		B9–5	Do you think the village has been revitalized because of the wind turbine?		Likert 5 scale
	Wind turbine disadvantage	B10–1	Do you think the wind turbine make noise?		Likert 5 scale
		B10–2	Do you think the wind turbine has damaged the original form of land?		Likert 5 scale
		B10–3	Do you think the wind turbine has damaged the landscape around it?		Likert 5 scale
		B10–4	Do you think the wind turbine has damaged animals and vegetation in the area where it is installed?		Likert 5 scale
		B10–5	Do you think the wind turbine causes psychological anxiety because of its installation?		Likert 5 scale
	Local acceptance	B11–1	Do you feel a stronger sense of belonging to the village after the wind turbine installation?	Park and Hwang (2016) [68]	Likert 5 scale
		B11–2	Do you think your interest in village development has been increased after the wind turbine installation?		Likert 5 scale
		B11–3	Do you think the village can be revitalized independently after the wind turbine installation?		Likert 5 scale
Level of environmental protection awareness	Perception of the importance of environmental issues	C12	What are the two most important environmental factors to you?	Kwak et al. (2016) [69]	Nominal
	Attitudes toward environmental issues	C13	How much do you agree with the statement: "Economic development is more important than environmental protection"?		Likert 5 scale
	Interests in environmental issues	C14	How interested are you in environmental issues in daily life?		Likert 5 scale
		C15	How interested are you in environmental protection?		Likert 5 scale
	Environmental practice	C16	How often do you engage in environmentally friendly behaviors?		Likert 5 scale
		C17	Have you changed your electricity consumption behavior after the wind turbine installation?		Likert 5 scale
Demographic characteristics		D1	Gender		Nominal
		D2	Age		Nominal
		D3	Residence year		Subjective
		D4	Village residence		Nominal
		D5	Household		Nominal
		D6	Education level		Nominal
		D7	Wind farm village member		Nominal
		D8	Job		Subjective

and regression analysis. The Kruskal-Wallis test was performed with multiple comparisons of villages, considering effect size. The regression analysis was conducted using a generalized linear model. We additionally conducted in-depth interviews with survey respondents to explore findings that were not fully captured through the surveys.

The questionnaire utilized in this study was meticulously designed to encompass inquiries concerning various aspects, including the types of benefit-sharing, the degree of local acceptance toward wind turbines, and the level of residents' environmental awareness. Additionally, demographic information such as age, duration of residence, type of residency, presence of a village committee, and occupation was incorporated into the questionnaire.

Structured closed-type items and optional questions formed the framework of the questionnaire. Data were collected through face-to-face interactions. Drawing from literature research and prior interviews, the structured questionnaire was constructed based on the Likert 5-point scale. The section of the questionnaire about local acceptance encompassed opinions before and after wind turbine installation, evaluations of the pros and cons of wind turbines, and assessments of village perception post-installation (See Table 3). Perception measures were collected on 5-point Likert scales. Specifically, X_1 (benefit-sharing) uses item A3, which asks respondents to rate the extent to which benefit-sharing is provided in their village, and X_2 (community engagement) uses item A5, which asks respondents to rate the extent of village-level participation in decision-making (e.g., public hearings/consultations, committee processes) as perceived by residents.

Consequently, higher scores indicate a more positive perception of advantages, whereas lower scores indicate a more negative view of disadvantages associated with wind turbines. Before fielding, the questionnaire was refined through preliminary interviews, expert consultation, and pilot testing to address ambiguities or errors. Given the study's focus on BSM, local acceptance, and environmental awareness, we assessed validity and reliability. Factor analysis with principal component extraction and orthogonal varimax rotation was used to identify latent structure. Factor retention followed the eigenvalue >1.0 criterion. Normality was examined using the Shapiro-Wilk test; results indicated *p*-values between 0.00 and 0.02 across all survey items, suggesting non-normal distributions.

3.5. Data analysis

Hypothesis 2-1 examines the level of awareness about wind turbines. Questions B9–1 to B9–5 ask whether wind turbines improve air quality, provide infinite energy sources, increase energy independence, offer economic benefits to the village, and contribute to village revitalization. All responses show positive scores from residents of Samdal-ri, Gasi-ri, and Haengwon-ri. Questions B10–1 to B10–5 assess perceptions of the disadvantages of wind turbines (noise, land damage, landscape damage, rand animal/vegetation damage, and psychological damage), using reverse-coded versions of the positive questions. The item was reverse-coded during data analysis, so lower scores indicate fewer negative effects, representing a more positive outcome.

Upon completion of data collection, the acquired data underwent rigorous analysis to assess validity and reliability, utilizing Python. First, given the non-normal distribution of all variables, the Kruskal-Wallis nonparametric test method was chosen, treating the dependent variable as continuous. Second, correlation analysis was conducted to explore potential relationships among the variables under consideration. Finally, multi-linear regression analysis was employed to examine associations among benefit-sharing, community engagement, local acceptance, and environmental awareness. For community engagement, the two binary indicators were summed to form index (0–2), rescaled to 0–1, and standardized (z-score) prior to modeling. This village-level operationalization focuses on decision-making participation and excludes ownership/equity to avoid construct conflation with benefit-sharing and to mitigate endogeneity concerns in a cross-sectional setting. Given the cross-sectional, non-randomized design, regression estimates are interpreted as associations conditional on included covariates and do not identify causal effects or their direction.

4. Results

4.1. Survey demographic characteristics

The questionnaire administered in this study sought to elucidate the demographic profile of respondents, encompassing variables such as gender, age, duration of residence, village birth status, membership status, and employment (See Table 4). Upon conducting demographic and sociological analyses, it was observed that respondents exhibited a balanced distribution across gender, age groups, and tenure of residence within the village. Many respondents identified as residents, representing household occupants. Village committee membership status distribution appeared relatively uniform, with conditions for committee membership often contingent upon ancestral lineage connections within the preceding committee.

4.2. Kruskal-Wallis test result

As shown in Table 5, we conducted the Kruskal–Wallis nonparametric test to examine Hypotheses 1–1, and 1–2, followed by post-hoc multiple comparisons to identify differences among the villages. The results revealed statistically significant differences ($p < 0.05$) in local acceptance levels (H1−1), whereas no significant differences were observed in environmental awareness (H1−2). Post-hoc analysis indicated that the most pronounced difference in local acceptance occurred between Gasi-ri and Haengwon-ri, with Gasi-ri showing notably higher scores. In contrast, Samdal-ri's scores did not significantly differ from those of either Gasi-ri or Haengwon-ri. This pattern suggests that local acceptance may be shaped not only by the presence of benefit-sharing or ownership schemes but also by contextual factors such as past experiences with wind projects, perceptions of fairness in benefit distribution, and the nature of interactions between developers and the community. Since these elements could not be fully captured through survey data alone, additional field visits and interviews with wind farm village committee leaders were conducted to explore these underlying factors in greater depth.

4.3. Correlation analysis result

The correlation analysis results are visualized in the heatmap shown in Fig. 8, enabling a clearer examination of the relationships among key variables. Particular attention was given to the link between perceived benefit-sharing (A3) and community engagement (A5), as well as their associations with awareness about wind turbines and environmental awareness. For Hypothesis 2-1, which tests whether perception of benefit-sharing is associated with local acceptance (B9-1 to B11-3), several correlations exceeded 0.3, indicating a moderate positive relationship. This suggests that higher perceptions of benefit-sharing tend to be accompanied by more favorable evaluations of wind turbines and their contribution to the village. For Hypothesis 2-2, which tests whether perception of benefit-sharing is related to environmental awareness (C13–C17), most correlations were negative and below zero. This pattern implies that stronger perceptions of benefit-sharing do not necessarily correspond to higher environmental awareness; in some cases, the association appears inverse. For Hypothesis 3–1, which examines whether community engagement is linked to local acceptance, the correlations between A5 and B9-1–B11-3 were generally small but positive, indicating a weak tendency for greater engagement to align with more favorable acceptance. However, for Hypothesis 3–2, the correlations between A5 and C13–C17 were close to zero, suggesting no meaningful relationship between community engagement and environmental awareness.

Table 4
Survey respondent characteristics.

		Sum		Samdal-ri		Gasi-ri		Haengwon-ri	
		Estimates (n = N)	Ratio (%)	Estimates (n = N)	Ratio (%)	Estimates (n = N)	Ratio (%)	Estimates (n = N)	Ratio (%)
Gender	Male	56	59	16	53	21	7	19	54
	Female	39	41	14	47	9	3	16	46
	Total	95	100	30	100	30	100	35	100
Age	30–39	6	6	0	0	1	3	5	14
	40–49	4	4	1	3	3	1	0	0
	50–59	27	28	16	53	3	1	8	23
	60–69	58	61	13	43	23	77	22	63
	Total	95	100	30	100	30	100	35	100
Years living in village	≤ 30	36	38	11	37	5	17	20	57
	> 30	59	62	19	63	25	83	15	43
	Total	95	100	30	100	30	100	35	100
Resident	Yes	94	99	30	100	30	100	34	97
	No	1	1	0	0	0	0	1	3
	Total	95	100	30	100	30	100	35	100
Household	Yes	59	62	20	67	21	70	18	51
	No	36	38	10	33	9	30	17	49
	Total	95	100	30	100	30	100	35	100
Education	Elementary	8	8	2	7	6	2	0	0
	Middle	17	18	1	3	9	3	7	2
	High school	40	42	17	57	7	23	16	46
	College	30	32	10	33	8	27	12	34
	Total	95	100	30	100	30	100	35	100
Village committee member	Yes	51	54	22	73	21	7	22	63
	No	44	46	8	27	9	3	13	37
	Total	95	100	30	100	30	100	35	100

Taken together, these findings underscore a stronger and more consistent connection between perceived benefit-sharing and local acceptance than between community engagement and either outcome. They also indicate that perception of benefit-sharing may influence local acceptance independently of its effect on environmental awareness. Consistent with Section 2.2.2., estimates for community engagement refer to decision-making participation, not ownership or equity.

4.4. Linear regression analysis result

To test Hypothesis 2, we conducted simple regression analyses focusing on the relationship between satisfaction with benefit sharing and local acceptance of wind turbines, as well as environmental awareness (see Table 6). The independent variable is residents' perception of benefit-sharing from wind turbines, measured by a 5-point Likert scale assessing satisfaction with the current benefit-sharing from the wind turbine in the questionnaire (A3). The dependent variables are each question from B9–1 to B11–3 and from C13 to C17. Questions on wind turbine disadvantages (B10–1 to B10–5) were reverse-coded during data processing, so all results from B9–1 to B11–3 are scaled such that higher scores indicate more positive perceptions of wind turbines.

As a result, most items in Hypothesis 2-1 displayed statistically significant positive associations with local acceptance, with the exception of B9–1 ("improve air quality"). Among the perceived advantages of wind turbines, beliefs about their role as an infinite energy source (B9–2), contribution to energy independence (B9–3), provision of economic benefits to the village (B9–4), and potential for village revitalization (B9–5) were all significant predictors. Similarly, all perceived disadvantages, such as noise, land damage, landscape damage, damage to local flora and fauna, and psychological impacts (B10–1 to B10–5), showed significant negative effects when reverse-coded, indicating that lower perceived harm is linked to higher acceptance. Notably, personal emotion variables (B11–1 to B11–3), reflecting a sense of belonging, interest in village development, and belief in the village's revitalization potential, emerged as particularly strong predictors. By contrast, none of the environmental awareness variables in Hypothesis 2-2 (C13 to C17) reached statistical significance, except for C14 ("interest in environmental problems in general"), which approached significance (p = 0.056). This pattern suggests that while benefit sharing and local engagement factors strongly influence acceptance, general environmental attitudes may play a less direct role.

Table 7 presents the regression analysis results for Hypothesis 3, which examined the effect of community engagement on local acceptance and environmental awareness. The independent variable is community engagement, measured by a 5-point Likert scale assessing residents' perceptions of preferred community engagement after experiencing current benefit-sharing and community engagement schemes in the village (A5). The dependent variables are each question from B9–1 to B11–3 and from C13 to C17. For Hypothesis 3–1, which focused on local acceptance, statistically significant associations ($p < 0.05$) were found for *wind turbine advantage - energy independence* (B9-3), *wind turbine disadvantage - noise* (B10-1), *wind turbine disadvantage - random animal and vegetaion damage* (B10-4), *personal emotion - interest in village development* (B11-2), and *personal emotion - revitalization power of the village* (B11-3). In contrast, no significant relationships were observed for Hypothesis 3–2 concerning environmental awareness.

When combining the regression analysis across all villages, both benefit-sharing and public hearings consistently influenced several key factors, including *wind turbine advantage - energy independence* (B9-3), *wind turbine disadvantage - noise* (B10-1), *wind turbine disadvantage - random animal and vegetation damage* (B10-4), *personal emotion - interest in village development* (B11-2), and *personal emotion - revitalization power of the village* (B11-3). Higher levels of benefit-sharing and community engagement were associated with stronger recognition of Jeju Island's potential for energy self-sufficiency, lower perceived noise impacts, reduced concerns about effects on livestock and vegetation, greater involvement in village affairs, and stronger aspirations for village revitalization. The effect of benefit-sharing appeared more pronounced than that of community engagement, as indicated by its consistent significance across additional variables, including *wind turbine advantage - infinite energy source* (B9-2), *wind turbine advantage - village revival* (B9-5), *wind turbine disadvantage - land damage* (B10-2), *wind turbine disadvantage - landscape daamge* (B10-3), *wind turbine disadvantage - psychological damage* (B10-5), and *personal emotion - sense of belonging to the village* (B11-1).

Table 5
Result of Kruskal-Wallis analysis between two villages.

Difference between two villages	Samdal-ri & Haengwon-ri		Samdal-ri & Gasi-ri		Gasi-ri & Haengwon-ri	
A. Awareness about benefit sharing						
Code	Chi-square statistics	P-value	Chi-square statistics	P-value	Chi-square statistics	P-value
A1	2.8	0.09	0.29	0.59	4.98	**0.03***
A2	0.26	0.61	19.2	**0***	10.81	**0***
A3	14.92	**0***	3.35	0.07	22.35	**0***
A4	3.23	0.07	4.79	**0.03***	2.16	0.14
A5	6.36	**0.01***	6.54	**0.01***	3.02	0.08
A6	4.16	**0.04***	3.19	0.07	0.03	0.86
B. Awareness about wind turbines						
Questions	Chi-square statistics	P-value	Chi-square statistics	P-value	Chi-square statistics	P-value
B7	2.42	0.12	4.54	**0.03***	1	0.32
B8	4.56	**0.03***	0.26	0.61	2.57	0.11
B9–1	0.12	0.73	2.11	0.15	3.75	0.05
B9–2	3.09	0.08	2.51	0.11	10.42	**0***
B9–3	2.7	0.1	4.47	**0.03***	12.97	**0***
B9–4	0.51	0.48	8.17	**0***	14.66	**0***
B9–5	0.24	0.63	8.05	**0***	10.41	**0***
B10–1	5.04	**0.02***	17.12	**0***	19.25	**0***
B10–2	2.37	0.12	7.82	**0.01***	11.53	**0***
B10–3	3.7	0.05	8.32	**0***	14.56	**0***
B10–4	1.67	0.2	16.29	**0***	13.3	**0***
B10–5	3.5	0.06	14.67	**0***	16.29	**0***
B11–1	6.79	**0.01***	14.06	**0***	27.48	**0***
B11–2	8.85	**0***	8.53	**0***	25.23	**0***
B11–3	13.63	**0***	4.56	**0.03***	25.44	**0***
C. Awareness about environmental issues						
Questions	Chi-square statistics	P-value	Chi-square statistics	P-value	Chi-square statistics	P-value
C12–1	0.65	0.42	0.14	0.71	0.1	0.76
C12–2	2.44	0.12	0.12	0.73	3.88	0.05
C13	0.02	0.9	0.06	0.81	0.04	0.84
C14	0.37	0.54	0.04	0.84	0.81	0.37
C15	2.21	0.14	1.2	0.27	0	0.95
C16	1.79	0.18	2.79	0.1	0.36	0.55
C17	0.73	0.39	1.14	0.29	0.08	0.78

* $p<0.05$

5. Discussion

Summarizing the validation outcomes for the hypotheses:

- **H1–1:** There are significant differences in local acceptance of wind power projects among residents in villages, thereby rejecting the null hypothesis.
- **H1–2:** There are no significant differences in environmental awareness among residents in villages, thereby supporting the null hypothesis.
- **H2–1**: Perception of benefit-sharing is positively associated with local acceptance, rejecting the null hypothesis.
- **H2–2**: Perception of benefit-sharing is not positively associated with environmental awareness, supporting the null hypothesis.
- **H3–1**: Community engagement is positively associated with local acceptance, rejecting the null hypothesis.
- **H3–2**: Community engagement is not positively associated with environmental awareness, supporting the null hypothesis.

Overall, the survey results indicate that financial benefit-sharing and community engagement significantly improve local acceptance of wind turbines, but neither approach leads to a measurable increase in environmental awareness (see Table 5). The relatively high levels of acceptance observed in Gasi-ri and Haengwon-ri, regardless of whether benefit-sharing is financially focused or engagement-oriented, align with the broader framework of the BSM [15,66]. However, the absence of significant differences in environmental awareness across Samdal-ri, Gasi-ri, and Haengwon-ri underscores a key limitation of BSM-focused approaches.

A comparison with the 2017 preliminary survey results further contextualizes the results for Hypothesis 1-1. According to Lee [70], which conducted the 2017 preliminary survey, the Kruskal-Wallis test of the same three villages revealed that benefit-focused model (Gasi-ri) and participation-focused model (Haengwon-ri) demonstrated higher local acceptance than utility-focused model (Samdal-ri). However, by 2024, disparities in local acceptance had emerged between Gasi-ri and Haewon-ri, with Gasi-ri showing a decline despite the continuation of benefit-sharing arrangements.

Regression analyses (see Tables 6 and 7) further indicate that both benefit-sharing and community engagement positively associate with perceptions of wind energy advantages, such as energy independence and village revitalization, while mitigating concerns such as noise and ecological impacts. Notably, benefit-sharing demonstrated a higher association than community engagement across multiple advantage and disadvantage indicators. Yet, neither variable had a significant effect on environmental awareness, reinforcing the need for an expanded framework that addresses this gap.

Post-survey interviews provided qualitative explanations for these patterns: First, in Gasi-ri, the land rent value embedded in benefit-sharing compensation did not keep pace with rising land prices after turbine installation, while land taxes for shared pasture increased, creating heavier tax burdens for residents, because of the long-term contracts for wind turbines (typically 10 or 15 years). These issues contributed to lower local acceptance in Gasi-ri between 2017 and 2024. Second, there is a lack of legal and technical support after installation of the wind turbines in villages. The onshore and offshore wind power plants are typically installed in rural villages, where older populations often work in agriculture or fishing. Although the government promotes renewable energy projects as part of a top-down national strategy, residents may interpret wind turbine installation differently, leading to declining satisfaction over time.

These findings underscore that financial benefits alone are insufficient to sustain long-term local acceptance of wind power projects and self-governance of village committee under the BSM. Given the typical 15 years operational life of wind turbines, inconsistencies in benefit-sharing implementation, combined with reduced financial returns and increased taxation, may erode support over time.

To address these limitations, we propose a shift toward a Value Sharing Mechanism (VSM), as a more comprehensive framework than the BSM. The VSM approach integrates environmental awareness into the existing BSM framework, emphasizing not only monetary compensation but also residents' collective role in sustainable energy transitions. The VSM not only considers benefit-sharing but also integrates community engagement based on environmental protection awareness. This approach incorporates community engagement practices grounded in environmental values, aligning with national energy and climate policy objectives.

The proposed VSM can also be anchored in the commons theory [16,18,20,21]. While Ostrom's eight design principles are not applied in a one-to-one mapping, this study draws on their core elements, such as clearly defined boundaries, locally tailored rules, participatory decision-making, and the right to self-organize, to frame community-based wind farms on shared pastureland. Prior studies have applied the commons framework to renewable energy, highlighting factors that foster community participation and trust ([20–22]). Our findings advance this literature by showing that even when commons-like governance structures exist, environmental awareness may not improve without intentional integration into benefit-sharing mechanisms. This extends the

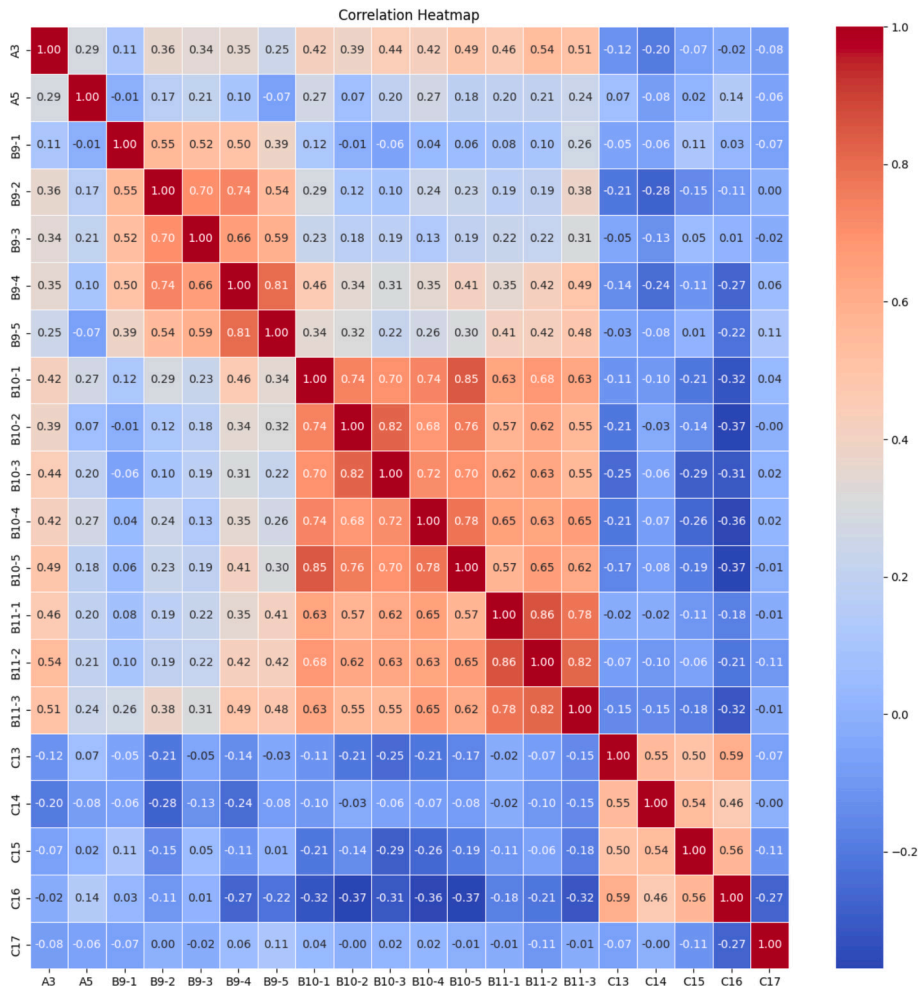

Fig. 8. Correlation heatmap.

Table 6
Result of Regression Analysis (Hypothesis 2).

Coefficients		Code	Estimate	Std. Error	t value	P-values
Hypothesis 2-1.						
Wind turbine advantage (B9)	improve air quality	B9–1	0.0987	0.093	1.065	0.289
	infinite energy source	B9–2	0.3864	0.104	3.715	**0.000***
	energy independence	B9–3	0.3053	0.088	3.474	**0.001***
	economic benefit to village	B9–4	0.4075	0.112	3.640	**0.000***
	village revival	B9–5	0.3128	0.128	2.454	**0.016***
Wind turbine disadvantage (B10)	noise	B10–1	0.5654	0.128	4.432	**0.000***
	land damage	B10–2	0.4884	0.121	4.035	**0.000***
	landscape damage	B10–3	0.5402	0.114	4.748	**0.000***
	rand animal/vegetation damage	B10–4	0.5331	0.118	4.510	**0.000***
	psychological damage	B10–5	0.6235	0.115	5.440	**0.000***
Personal emotion (B11)	sense of belonging to village	B11–1	0.5664	0.114	4.986	**0.000***
	interest in village development	B11–2	0.6912	0.113	6.108	**0.000***
	revitalization power of village	B11–3	0.6418	0.113	5.671	**0.000***
Hypothesis 2-2.						
Agreement on importance of economic development than environmental protection		C13	−0.0923	0.081	−1.142	0.257
Interest in environmental problems in general		C14	−0.1609	0.083	−1.937	0.056
Importance of environmental protection		C15	0.0921	0.174	0.528	0.599
Behavior for environmental protection		C16	−0.0464	0.073	−0.633	0.529
Behavior changes in electricity consumption		C17	−0.0173	0.091	−0.190	0.850

* $p<0.05$

commons perspective beyond financial arrangements, toward a hybrid model that embeds environmental values in governance practices.

From this perspective, the VSM should incorporate accessible conflict resolution, participatory decision-making, and nested larger networks among the windfarm villages. Policy implications include: (1) setting lease fees that reflect future monetary values of shared

Table 7
Result of Regression Analysis (Hypothesis 3).

Coefficients		Code	Estimate	Std. Error	t value	P-values
Hypothesis 3–1.						
Wind turbine advantage (B9)	improve air quality	B9–1	−0.0041	0.040	−0.105	0.917
	infinite energy source	B9–2	0.0758	0.047	1.627	0.107
	energy independence	B9–3	0.0808	0.039	2.087	**0.040***
	economic benefit to village	B9–4	0.0493	0.051	0.976	0.332
	village revival	B9–5	−0.0379	0.056	−0.681	0.497
Wind turbine disadvantage (B10)	noise	B10–1	0.1527	0.057	2.660	**0.009***
	land damage	B10–2	0.0352	0.056	0.635	0.527
	landscape damage	B10–3	0.1014	0.053	1.923	0.058
	rand animal/vegetation damage	B10–4	0.1427	0.053	2.675	**0.009***
	psychological damage	B10–5	0.0983	0.055	1.791	0.077
Personal emotion (B11)	sense of belonging to village	B11–1	0.1032	0.053	1.941	0.055
	interest in village development	B11–2	0.1132	0.056	2.036	**0.045***
	revitalization power of village	B11–3	0.1264	0.054	2.335	**0.022***
Hypothesis 3–2.						
Agreement on importance of economic development than environmental protection		C13	0.0224	0.034	0.651	0.517
Interest in environmental problems in general		C14	−0.0270	0.036	−0.753	0.454
Importance of environmental protection		C15	0.0052	0.031	0.167	0.868
Behavior for environmental protection		C16	0.0521	0.038	1.358	0.178
Behavior changes in electricity consumption		C17	−0.0164	0.030	−0.540	0.590

* $p<0.05$

pastureland that reduce residents' financial burdens; (2) providing technical support for facility operation, management, and maintenance to enhance the long-term performance of wind power generation systems; (3) offering legal and tax advisory assistance to help address administrative and financial complexities; and (4) mandating public hearings and implementing environmental education programs as integral components of wind power development plans to strengthen environmental awareness.

In case of Jeju Island, these results can inform renewable energy policies under the 2 Ordinance on Conservation Area Management in Jeju Special Self-Governing Province. Consistent with the Carbon Free Island by 2030 roadmap, the Gasi-ri Wind Power Construction Plan could integrate provisions for long-term contracts, tax relief, and enhanced public hearings that link installation purposes with environmental awareness initiatives. Such measures would strengthen both the cultural and material dimensions of the island's energy transition, as emphasized in Cho and Jeon [71].

6. Conclusion

This study empirically analyzed whether BSM is associated with local acceptance and environmental awareness in the context of wind power projects. The findings indicate that both perception of benefit-sharing and community participation are positively associated with higher local acceptance, but not with environmental awareness. This gap highlights a critical limitation of current BSM-based approaches and points to the need for a more comprehensive framework.

Drawing on these results, we propose the VSM as an expanded model that integrates environmental awareness into the existing BSM structure. By incorporating community engagement strategies that emphasize environmental stewardship, the VSM can foster a shared understanding among residents of their contribution to sustainable energy systems. Such an approach is not only aligned with the national government's broader environmental and energy policy objectives, but also offers a pathway for strengthening long-term self-governance of community-based renewable energy projects.

From a commons theory perspective ([19–21]), this study frames wind resources on shared pastureland as common-pool resources. The results underscore the importance of governance principles, such as clearly defined boundaries, locally tailored rules, participatory decision-making, and accessible conflict resolution, in sustaining local acceptance over the long term. Applying these principles to VSM design suggests policy implications such as adjusting lease fees to reflect future land values, providing technical and maintenance support, offering legal and tax assistance, and mandating public hearings and environmental education. These measures can enhance both the cultural and material dimensions of energy transition, as evidenced in the Jeju Island case.

However, this study has several limitations. First, the VSM framework was tested with a limited set of variables, benefit-sharing, community engagement, local acceptance, and environmental awareness, and future research should incorporate additional socioeconomic, institutional, and environmental factors. Second, the analysis was confined to selected onshore and offshore wind farms villages in Jeju; applying the framework to other geographic contexts could provide broader insights. Third, while this study primarily reflects the perspectives of residents, future research should capture the interests and power dynamics among multiple stakeholders, including developers, policymakers, and environmental groups. Because the data are cross-sectional, the direction of influence cannot be established; reverse causality and omitted-variable bias may remain. Findings should be treated as hypothesis-generating rather than confirmatory.

Future research should also evaluate the VSM framework longitudinally to examine how local acceptance and environmental awareness evolve over time, and to identify potential tipping points for sustained community support. Comparative studies across multiple renewable energy technologies, such as solar, tidal, and biomass, would help assess whether commons-based governance principles operate differently under varying technological, spatial, and cultural conditions. By extending the scope of analysis and integrating cross-case learning, researchers can refine the VSM into a generalizable model for community-led renewable energy governance worldwide.

Overall, by combining empirical evidence with the commons theory framework, this study advances the understanding of how value-sharing arrangements can be designed to balance economic benefits, environmental awareness, and community self-governance in the energy transition.

CRediT authorship contribution statement

Kyungmin Lee: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Sun-Jin Yun:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization. **Ji Yong Myeong:** Writing – review & editing, Visualization, Validation,

Investigation, Formal analysis, Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] IEA, Korea 2020. 2020. International Energy Agency: Paris, Retrieved from, https://www.iea.org/reports/korea-2020, 2020. Licence: CC BY 4.0.
[2] Framework Act on Carbon Neutrality and Green Growth for Coping with Climate Crisis, Act No. 18469, 2021. https://elaw.klri.re.kr/eng_mobile/viewer.do?hseq=68875&type=part&key=39.
[3] IEA, Strategies for Coal Transition in Korea. 2023, International Energy Agency, Paris, 2023. Retrieved from, https://www.iea.org/reports/strategies-for-coal-transition-in-korea. Licence: CC BY 4.0.
[4] Korea Ministry of Trade, Industry, and Energy, The 10th Basic Plan for Long-Term Electricity Supply and Demand (2022–2036), *Korea Ministry of Trade, Industry, and Energy*, Sejong, Korea, 2023, 2023. Retrieved from, https://www.motie.go.kr/kor/article/ATCLc01b2801b/68162/view#.
[5] J. Park, B. Kim, (2019) An analysis of South Korea's energy transition policy with regards to offshore wind power development, Renew. Sust. Energ. Rev. 1 (109) (2019) 71–84, https://doi.org/10.1016/j.rser.2019.04.031. Jul.
[6] J. Ki, S.J. Yun, W.C. Kim, S. Oh, J. Ha, E. Hwangbo, H. Lee, S. Shin, S. Yoon, H. Youn, (2022) Local residents' attitudes about wind farms and associated noise annoyance in South Korea, Energy Policy 1 (163) (2022), https://doi.org/10.1016/j.enpol.2022.112847, 112847. Apr.
[7] H. Kim, S.H. Cho, S. Song, (2019) Wind, power, and the situatedness of community engagement, Public Underst. Sci. 28 (1) (2019) 38–52, https://doi.org/10.1177/0963662518772508. Jan.
[8] S. Park, S.J. Yun, K. Cho, (2022) Public dialogue as a collaborative planning process for offshore wind energy projects: Implications from a text analysis of a South Korean case, Renew. Sust. Energ. Rev. 1 (169) (2022), https://doi.org/10.1016/j.rser.2022.112949, 112949. Nov.
[9] J. Choi, J.J. Kim, J. Lee, (2024) Public willingness to pay for mitigating local conflicts over the construction of renewable energy facilities: A contingent valuation study in South Korea, Energy Policy 1 (185) (2024), https://doi.org/10.1016/j.enpol.2023.113930, 113930. Feb.
[10] J. Lomax, N. Mirumachi, M. Hautsch, (2023) Does renewable energy affect violent conflict? Exploring social opposition and injustice in the struggle over the Lake Turkana Wind Farm, Kenya, Energy Res. Soc. Sci. 1 (100) (2023), https://doi.org/10.1016/j.erss.2023.103089, 103089. Jun.
[11] K.M. Lee, S.J. Yun, (2018) Positive effects and problematic conditions of adopting the benefit sharing mechanism in renewable energy projects: Focusing on the case of onshore wind farms in Jeju Island, Space and Environment. 28 (3) (2018) 227–270, https://doi.org/10.19097/kaser.2018.28.3.227.
[12] E.C. Lee, B.W. Lee, H.M. Kim, (2022) Strategies for the symbiotic development of renewable power and nuclear power under the 'Renewable Energy 3020' policy in South Korea: Part II. Technical solutions for expanding renewable power generation, Energy Rep. 8 (2022) 630–646, https://doi.org/10.1016/j.egyr.2022.10.227. Dec 1.
[13] E. van de Grift, E. Cuppen, (2022) Beyond the public in controversies: A systematic review on social opposition and renewable energy actors, Energy Res. Soc. Sci. 1 (91) (2022), https://doi.org/10.1016/j.erss.2022.102749, 102749. Sep.
[14] S. Kerr, K. Johnson, S. Weir, (2017) Understanding community benefit payments from renewable energy development, Energy Policy 1 (105) (2017) 202–211, https://doi.org/10.1016/j.enpol.2017.02.034. Jun.
[15] F.D. Musall, O. Kuik, (2011) Local acceptance of renewable energy—A case study from southeast Germany, Energy Policy 39 (6) (2011) 3252–3260, https://doi.org/10.1016/j.enpol.2011.03.017. Jun 1.
[16] D. Rudolph, C. Haggett, M. Aitken, (2018) Community benefits from offshore renewables: The relationship between different understandings of impact, community, and benefit, Environment and Planning C: Politics and Space. 36 (1) (2018) 92–117, https://doi.org/10.1177/2399654417699206. Feb.
[17] J.C. Rogers, E.A. Simmons, I. Convery, A. Weatherall, (2008) Public perceptions of opportunities for community-based renewable energy projects, Energy Policy 36 (11) (2008) 4217–4226, https://doi.org/10.1016/j.enpol.2008.07.028. Nov 1.
[18] G. Hardin, The tragedy of the commons: the population problem has no technical solution; it requires a fundamental extension in morality, Science 162 (3859) (1968) 1243–1248, https://doi.org/10.1126/science.162.3859.1243, 1968 Dec 13.
[19] E. Ostrom, (1990) Governing the commons: The evolution of institutions for collective action, Cambridge University Press, 1990. Retrieved from, https://digitalrepository.unm.edu/nrj/vol32/iss2/6.
[20] E. Ostrom, (1999) Coping with tragedies of the commons, Annu. Rev. Polit. Sci. 2 (1) (1999) 493–535, https://doi.org/10.1146/annurev.polisci.2.1.493. Jun.
[21] E. Ostrom, (2002) Common-pool resources and institutions: Toward a revised theory, Handb. Agric. Econ. 1 (2) (2002) 1315–1339, https://doi.org/10.1016/S1574-0072(02)10006-5. Jan.
[22] M. Cox, G. Arnold, S.V. Tomás, A review of design principles for community-based natural resource management, Ecol. Soc. 15 (4) (2010), 2010 Dec 1. Doi:https://www.jstor.org/stable/26268233.
[23] T. Bauwens, B. Gotchev, L. Holstenkamp, (2016) What drives the development of community energy in Europe? The case of wind power cooperatives, Energy Res. Soc. Sci. 1 (13) (2016) 136–147, https://doi.org/10.1016/j.erss.2015.12.016. Mar.
[24] R. Wade, G. Ellis, Reclaiming the Windy Commons: Landownership, Wind Rights, and the Assetization of Renewable Resources, Energies 15 (2022) 3744, https://doi.org/10.3390/en15103744.
[25] K. Andersson, Understanding decentralized forest governance: an application of the institutional analysis and development framework, Sustain. Sci. Pract. Policy 2 (1) (2006) 25–35, https://doi.org/10.1080/15487733.2006.11907975.
[26] M. Munday, G. Bristow, R. Cowell, (2011) Wind farms in rural areas: How far do community benefits from wind farms represent a local economic development opportunity? J. Rural. Stud. 27 (1) (2011) 1–2, https://doi.org/10.1016/j.jrurstud.2010.08.003. Jan 1.
[27] N. Cass, G. Walker, P. Devine-Wright, (2010) Good neighbours, public relations and bribes: the politics and perceptions of community benefit provision in renewable energy development in the UK, J. Environ. Policy Plan. 12 (3) (2010) 255–275, https://doi.org/10.1080/1523908X.2010.509558. Sep 1.
[28] B. Slee, (2015) Is there a case for community-based equity participation in Scottish on-shore wind energy production? Gaps in evidence and research needs, Renew. Sust. Energ. Rev. 1 (41) (2015) 540–549, https://doi.org/10.1016/j.rser.2014.08.064. Jan.
[29] C. Macdonald, J. Glass, E. Creamer, (2017) What is the benefit of community benefits? Exploring local perceptions of the provision of community benefits from a commercial wind energy project, Scott. Geogr. J. 133 (3–4) (2017) 172–191, https://doi.org/10.1080/14702541.2017.1406132. Oct 2.
[30] T. Ejdemo, P. Söderholm, (2015) Wind power, regional development and benefit-sharing: The case of Northern Sweden, Renew. Sust. Energ. Rev. 47 (2015) 476–485, https://doi.org/10.1016/j.rser.2015.03.082Get rights and content. Jul 1.
[31] P. Enevoldsen, B.K. Sovacool, (2016) Examining the social acceptance of wind energy: Practical guidelines for onshore wind project development in France, Renew. Sust. Energ. Rev. 1 (53) (2016) 178–184, https://doi.org/10.1016/j.rser.2015.08.041. Jan.
[32] H.J. Kooij, M. Oteman, S. Veenman, K. Sperling, D. Magnusson, J. Palm, F. Hvelplund, (2018) Between grassroots and treetops: Community power and institutional dependence in the renewable energy sector in Denmark, Sweden and the Netherlands, Energy Res. Soc. Sci. 1 (37) (2018) 52–64, https://doi.org/10.1016/j.erss.2017.09.019. Mar.
[33] L. Gorroño-Albizu, K. Sperling, S. Djørup, (2019) The past, present and uncertain future of community energy in Denmark: Critically reviewing and conceptualising citizen ownership, Energy Res. Soc. Sci. 1 (57) (2019), https://doi.org/10.1016/j.erss.2019.101231, 101231. Nov.
[34] K. Langer, T. Decker, J. Roosen, K. Menrad, (2018) Factors influencing citizens' acceptance and non-acceptance of wind energy in Germany, J. Clean. Prod. 20 (175) (2018) 133–144, https://doi.org/10.1016/j.jclepro.2017.11.221. Feb.
[35] IEA-RETD, Cost and financing aspects of community renewable energy projects, in: Volume I: Main Report. Ricardo Energy & Environment, IEA-RETD Operating Agent, IEA Implementing Agreement for Renewable Energy Technology Deployment (IEA-RETD), 2016, p. 2016. Utrecht.
[36] M.B. Roberts, A. Sharma, I. MacGill, (2022) Efficient, effective and fair allocation of costs and benefits in residential energy communities deploying shared photovoltaics, Appl. Energy 1 (305) (2022), https://doi.org/10.1016/j.apenergy.2021.117935, 117935. Jan.
[37] S.M. Hammami, A. Triki, Identifying the determinants of community acceptance of renewable energy technologies: The case study of a wind energy project from Tunisia, Renew. Sust. Energ. Rev. 1 (54) (2016) 151–160, https://doi.org/10.1016/j.rser.2015.09.037. Feb.
[38] J. Woo, S. Chung, C.Y. Lee, S.Y. Huh, (2019) Willingness to participate in community-based renewable energy projects: A contingent valuation study in South Korea, Renew. Sust. Energ. Rev. 1 (112) (2019) 643–652, https://doi.org/10.1016/j.rser.2019.06.010. Sep.
[39] B.K. Sovacool, (2013) The avian benefits of wind energy: A 2009 update, Renew. Energy 1 (49) (2013) 19–24, https://doi.org/10.1016/j.renene.2012.01.074. Jan.
[40] F. Rossi, M. Heleno, R. Basosi, A. Sinicropi, (2021) LCA driven solar compensation mechanism for Renewable Energy Communities: The Italian case, Energy 15 (235) (2021), https://doi.org/10.1016/j.energy.2021.121374, 121374. Nov.
[41] S. Trandafir, P. Thomas, D. Bidwell, R. Rezendes, (2023) Community benefit agreements for solar energy: Examining values, preferences and perceived benefits in the United States using a discrete choice experiment, Energy Res. Soc. Sci. 1 (106) (2023), https://doi.org/10.1016/j.erss.2023.103305, 103305. Dec.

[42] C. Schulz, J. Skinner, (2021) Hydropower benefit-sharing and resettlement: A conceptual review, Energy Res. Soc. Sci. 1 (83) (2022), https://doi.org/10.1016/j.erss.2021.102342, 102342. Jan.
[43] B.K. Sovacool, P.L. Ratan, (2012) Conceptualizing the acceptance of wind and solar electricity, Renew. Sust. Energ. Rev. 16 (7) (2012) 5268–5279, https://doi.org/10.1016/j.rser.2012.04.048. Sep 1.
[44] C.R. Warren, M. McFadyen, Does community ownership affect public attitudes to wind energy? A case study from south-west Scotland, Land Use Policy 27 (2) (2010) 204–213, https://doi.org/10.1016/j.landusepol.2008.12.010.
[45] S. Ahn, Y. Soh, H. Ryu, H. Lee, E. Hwangbo, S.J. Yun, (2021) An analysis of policy initiatives and benefit sharing schemes to support floating solar power plants, New & Renewable Energy. 17 (4) (2021) 9–27, https://doi.org/10.7849/ksnre.2021.0023.
[46] B.K. Sovacool, P. Enevoldsen, C. Koch, R.J. Barthelmie, (2017) Cost performance and risk in the construction of offshore and onshore wind farms, Wind Energy. 20 (5) (2017) 891–908, https://doi.org/10.1002/we.2069. May.
[47] M. Aitken, (2010) Wind power and community benefits: Challenges and opportunities, Energy Policy 38 (10) (2010) 6066–6075, https://doi.org/10.1016/j.enpol.2010.05.062. Oct 1.
[48] F.D. Minuto, A. Lanzini, Energy-sharing mechanisms for energy community members under different asset ownership schemes and user demand profiles, Renew. Sust. Energ. Rev. 168 (2022) 112859, https://doi.org/10.1016/j.rser.2022.112859.
[49] Action Renewables; Centre for Sustainable Energy & Ricardo Aea, Communities and Renewable Energy: A Study. Report commissioned by DETI, DoE and DARD, 2015, 2015 April. Retrieved from, https://www.economy-ni.gov.uk/publications/communities-and-renewable-energy-study.
[51] J. Knauf, (2022) Can't buy me acceptance? Financial benefits for wind energy projects in Germany, Energy Policy 1 (165) (2022), https://doi.org/10.1016/j.enpol.2022.112924, 112924. Jun.
[52] J. van Wijk, I. Fischhendler, G. Rosen, L. Herman, (2021) Penny wise or pound foolish? Compensation schemes and the attainment of community acceptance in renewable energy, Energy Res. Soc. Sci. 1 (81) (2021), https://doi.org/10.1016/j.erss.2021.102260, 102260. Nov.
[53] K. Reilly, A.M. O'Hagan, G. Dalton, (2016) Developing benefit schemes and financial compensation measures for fishermen impacted by marine renewable energy projects, Energy Policy 1 (97) (2016) 161–170, https://doi.org/10.1016/j.enpol.2016.07.034. Oct.
[54] G. Allan, P. Mcgregor, K. Swales, (2011) The importance of revenue sharing for the local economic impacts of a renewable energy project: A social accounting matrix approach, Reg. Stud. 45 (9) (2011) 1171–1186, https://doi.org/10.1080/00343404.2010.497132. Oct 1.
[55] S. Abolhosseini, A. Heshmati, (2014) The main support mechanisms to finance renewable energy development, Renew. Sust. Energ. Rev. 1 (40) (2014) 876–885, https://doi.org/10.1016/j.rser.2014.08.013. Dec.
[56] F. Goedkoop, P. Devine-Wright, (2016) Partnership or placation? The role of trust and justice in the shared ownership of renewable energy projects, Energy Res. Soc. Sci. 1 (17) (2016) 135–146, https://doi.org/10.1016/j.erss.2016.04.021. Jul.
[57] T. Howard, (2015) Olivebranches and idiot's guides: Frameworks for community engagement in Australian wind farm development, Energy Policy 1 (78) (2015) 137–147, https://doi.org/10.1016/j.enpol.2014.12.026. Mar.
[58] S.H. Lee, S.G. Yun, (2015) Review of measures to enhance local acceptance of renewable energy projects. Environmental Law and Policy 15, 2015, pp. 133–166, null. Retrieved from: https://www.dbpia.co.kr/Journal/articleDetail?nodeId=NODE06521959.
[59] L.T. Clausen, D. Rudolph, (2020) Renewable energy for sustainable rural development: Synergies and mismatches, Energy Policy 1 (138) (2020), https://doi.org/10.1016/j.enpol.2020.111289, 111289. Mar.
[60] S.H. Lee, S.G. Yun, Review of measures to enhance local acceptance of renewable energy projects, Environmental Law and Policy 15 (null) (2015) 133–166, https://doi.org/10.18215/envlp.15.201509.15.133.
[61] R. Wüstenhagen, M. Wolsink, M.J. Bürer, (2007) Social acceptance of renewable energy innovation: An introduction to the concept, Energy Policy 35 (5) (2007) 2683–2691, https://doi.org/10.1016/j.enpol.2006.12.001. May 1.
[62] M.A. Petrova, (2013) NIMBYism revisited: public acceptance of wind energy in the United States, Wiley Interdiscip. Rev. Clim. Chang. 4 (6) (2013) 575–601, https://doi.org/10.1002/wcc.250. Nov.
[63] M.A. Petrova, (2016) From NIMBY to acceptance: Toward a novel framework—VESPA—For organizing and interpreting community concerns, Renew. Energy 1 (86) (2016) 1280–1294, https://doi.org/10.1016/j.renene.2015.09.047. Feb.
[64] A. Dugstad, K. Grimsrud, G. Kipperberg, H. Lindhjem, S. Navrud, (2020) Acceptance of wind power development and exposure–Not-in-anybody's-backyard, Energy Policy 1 (147) (2020), https://doi.org/10.1016/j.enpol.2020.111780, 111780. Dec.
[65] S. Park, S.J. Yun, K. Cho, (2024) Energy justice: Lessons from offshore wind farm siting conflicts in South Korea, Energy Policy 1 (185) (2024), https://doi.org/10.1016/j.enpol.2023.113972, 113972. Feb.
[66] G. Walker, P. Devine-Wright, Community renewable energy: What should it mean? Energy Policy 36 (2) (2008) 497–500, https://doi.org/10.1016/j.enpol.2007.10.019.
[67] A.L. Berka, E. Creamer, Taking stock of the local impacts of community owned renewable energy: A review and research agenda, Renew. Sust. Energ. Rev. 82 (2018) 3400–3419, https://doi.org/10.1016/j.rser.2017.10.050.
[68] J. Park, S. Hwang, Study of the Effect of Visibility on the Acceptance of Southwest 2.5GW Offshore Wind Power. New & Renewable, Energy 12 (3) (2016) 13–19, https://doi.org/10.7849/ksnre.2016.9.12.3.13.
[69] S. Kwak, C. Lee, Y. Im, Public Attitudes Towards the Environment–2016 Survey, in: Basic Research Report 2016, Seoul: Korea Environment Institute, 2016, pp. 1–249. UCI: I410-ECN-0102-2018-500-000487538.
[70] K.M. Lee, (2017) An Influence of the Benefit Sharing Mechanism to the Local Acceptance and Environmental Awareness: A Case Study from Onshore Wind Farms in Jeju Island, Seoul National University. South Korea, Seoul, 2017. Retrieved from: https://s-space.snu.ac.kr/bitstream/10371/142605/1/000000150335.pdf.
[71] S.H. Cho, C. Jeon, Implanting Wind Turbines: Cultural and Material Dimensions of Energy Transition in Jeju, South Korea, East Asian Sci. Technol. Soc. 1-20 (2025), https://doi.org/10.1080/18752160.2025.2483103.